

ECON 2: Principles of Macroeconomics

Lecture 13: Deficits, Debt, and Monetary Policy



2009: American Recovery and Reinvestment Act (ARRA) Economic Stimulus Bill

- Signed into Law: February 17, 2009, by President Barack Obama.
- Key Figures
- Original Budget: \$787 billion (later revised to \$831 billion) :
 - \$288 billion in tax cuts and incentives.
 - \$224 billion for entitlement programs (e.g., unemployment benefits, Medicaid).
 - \$275 billion for federal contracts, grants, and loans
- Major Spending Areas
 - Infrastructure: \$85 billion for projects like roads, bridges, and mass transit
 - Healthcare: \$155 billion for Medicaid expansion, COBRA subsidies, and health IT.
 - Education: \$100 billion for schools and state fiscal stabilization.
 - Energy: \$90 billion for renewable energy projects and efficiency upgrades.

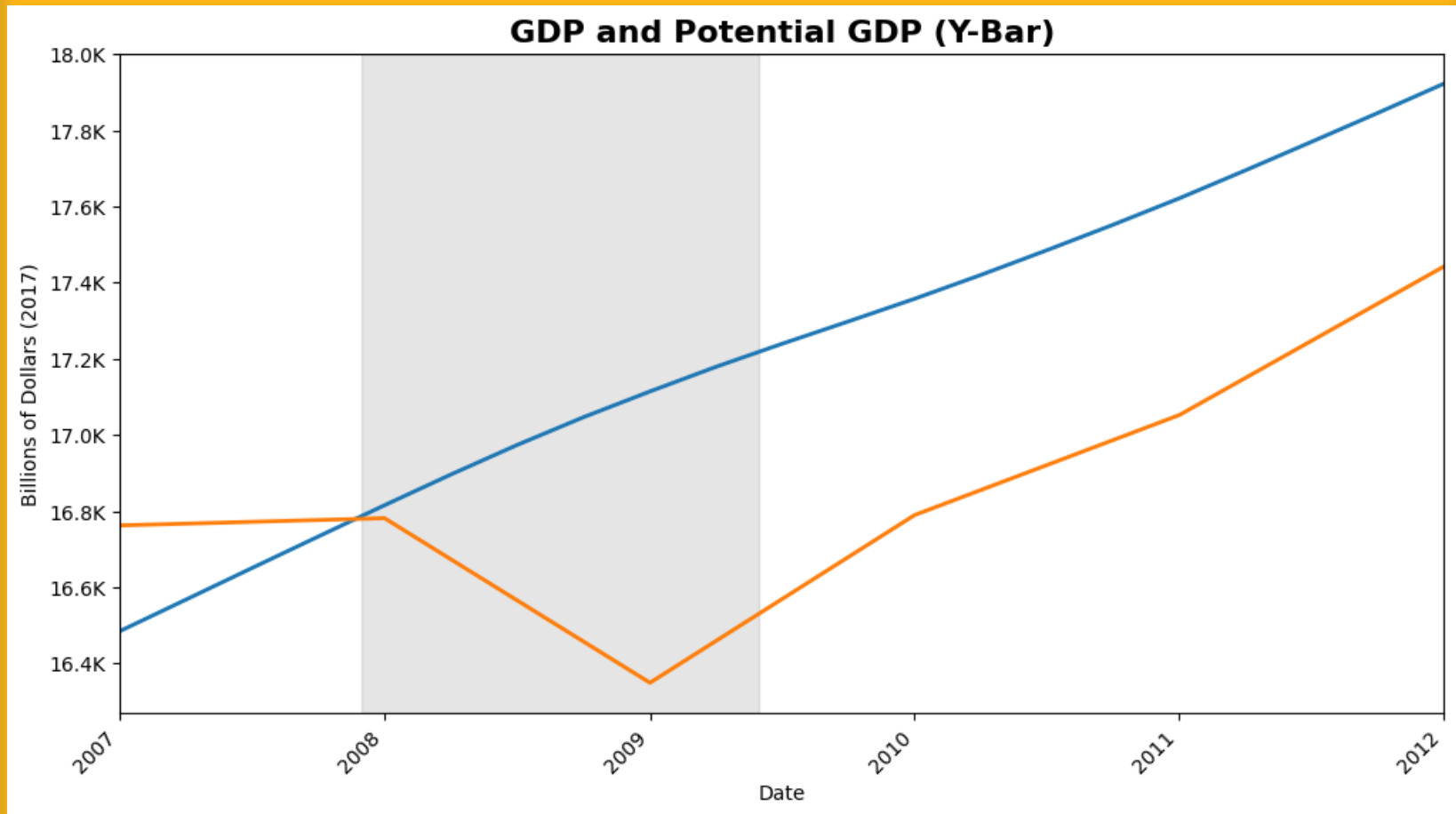


2009: American Recovery and Reinvestment Act (ARRA) Economic Stimulus Bill

- Economic Impact
 - Boosted GDP growth by up to 4.5% in 2010.
 - Created or saved an estimated 1.4–3.3 million jobs by mid-2010.
 - Helped lower unemployment by up to 1.8 percentage points.
- Criticisms
 - Spending delays hindered immediate economic recovery (“no such thing as shovel-ready projects”).
 - Concerns over waste, fraud, and inefficiency
 - Critics argued it increased federal debt without sufficient long-term benefits.
- Legacy
 - ARRA played a significant role in stabilizing the U.S. economy during the recession but remains controversial regarding its effectiveness and cost.

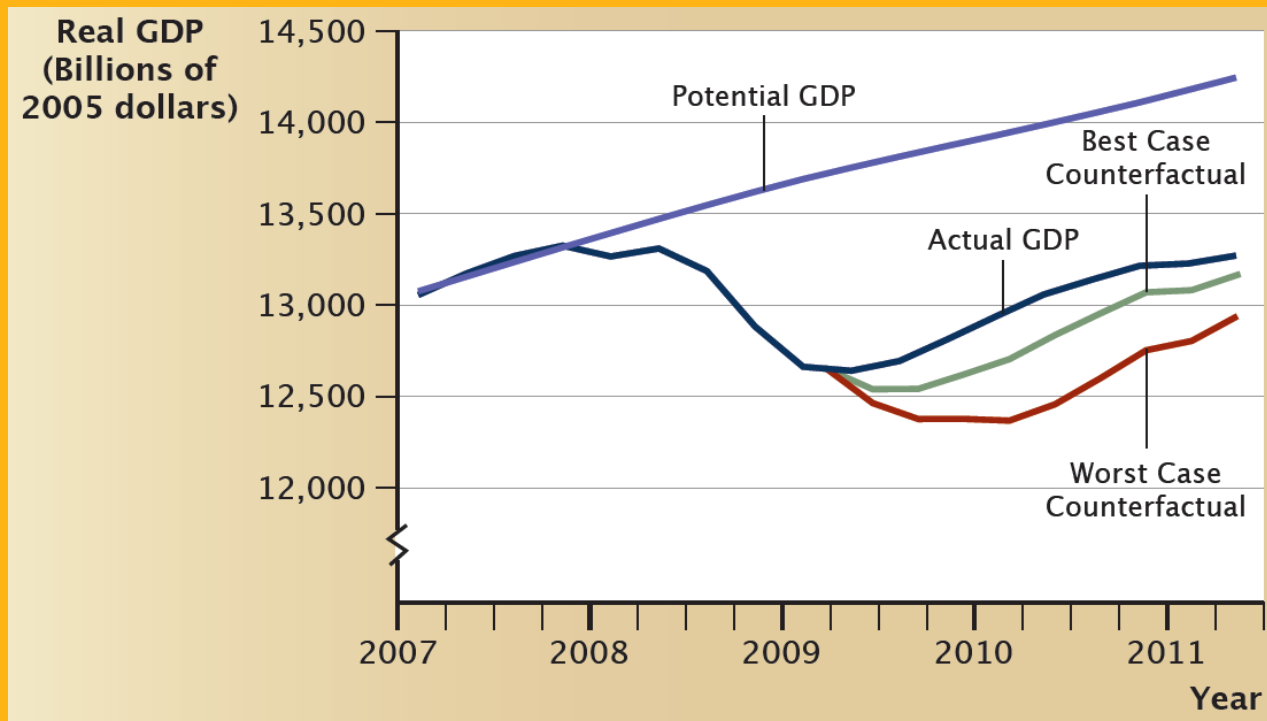


Government Response to the Great Recession



Government Response to the Great Recession

- Was stimulus useless?



Government Response to the Great Recession

Purchases of goods and services by the federal government	1.0 to 2.5
Transfers to state and local governments for infrastructure	0.5 to 2.5
Transfers to state and local governments for other purposes	0.7 to 1.8
Transfer payments to individuals (such as unemployment benefits and food stamps)	0.8 to 2.1
One-time payments to retirees and disabled individuals	0.3 to 1.0
One-time payments to existing recipients of Social Security	0.3 to 1.0
Making Work Pay tax credit	0.5 to 1.7
AMT patch	0.2 to 0.6
Other individual tax provisions	0.3 to 1.5

Government Response to the Great Recession

The size of the Great Recession stimulus package is still debated

Some say it was too small to stimulate the economy

Others claim it was too large and led to inflation and debt

Problems with the package's design besides its size



Covid-19 Pandemic Fiscal Policy

CARES Act (March 27, 2020)

- **Total Cost: \$2.2 trillion (largest in history)**

Key Provisions:

- \$290 billion in direct payments to individuals
- \$260 billion for unemployment benefits
- \$377 billion in small business relief
- \$150 billion for state and local governments
- \$200 billion for hospitals and healthcare providers.



Consolidated Appropriations Act (Dec. 2020)

- **Total Cost: \$900 billion**

Key Provisions:

- \$119 billion for unemployment benefits
- \$302 billion for small businesses
- \$82 billion for education

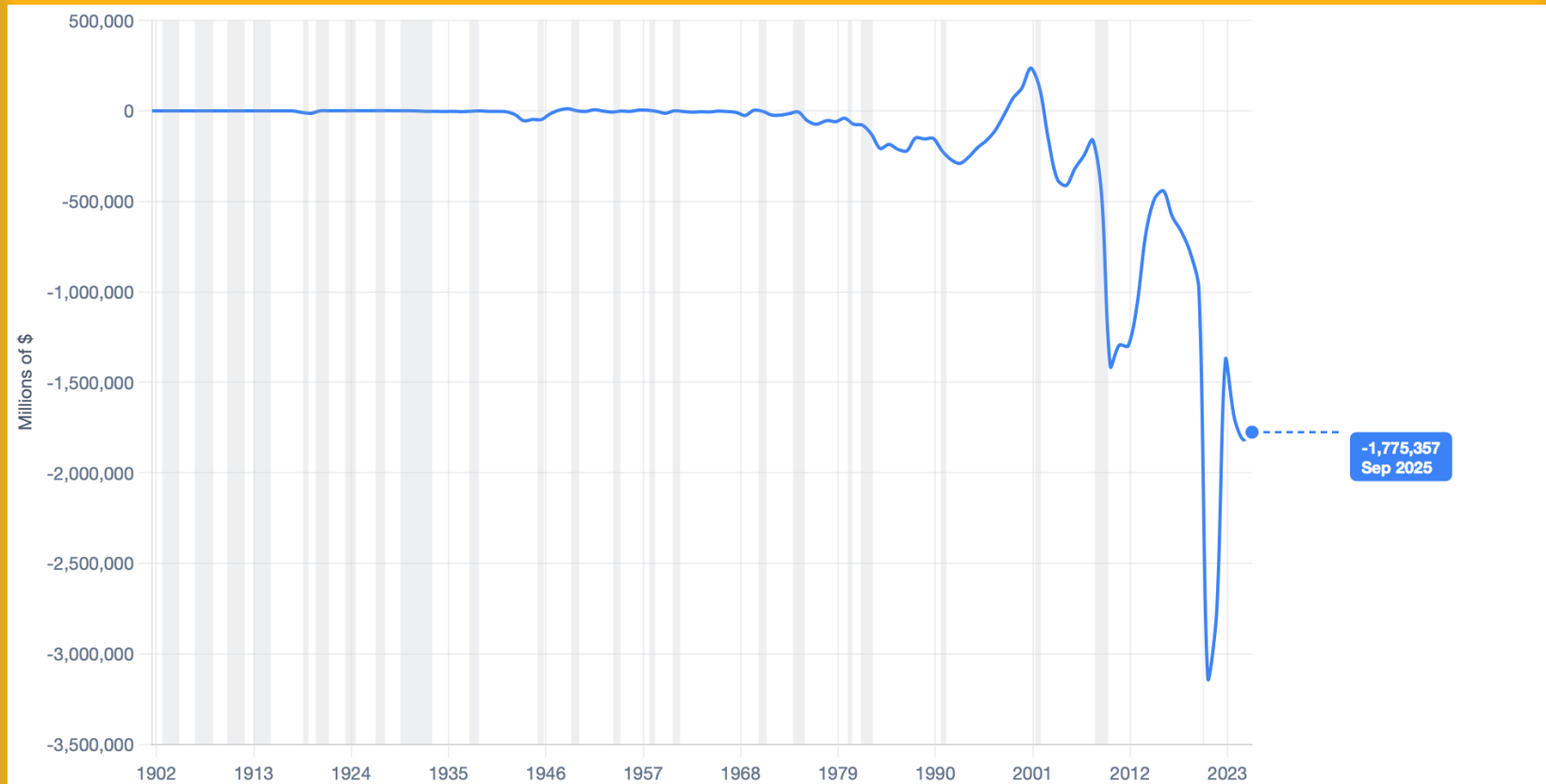
American Rescue Plan Act (March 2021)

- **Total Cost: \$1.9 trillion**

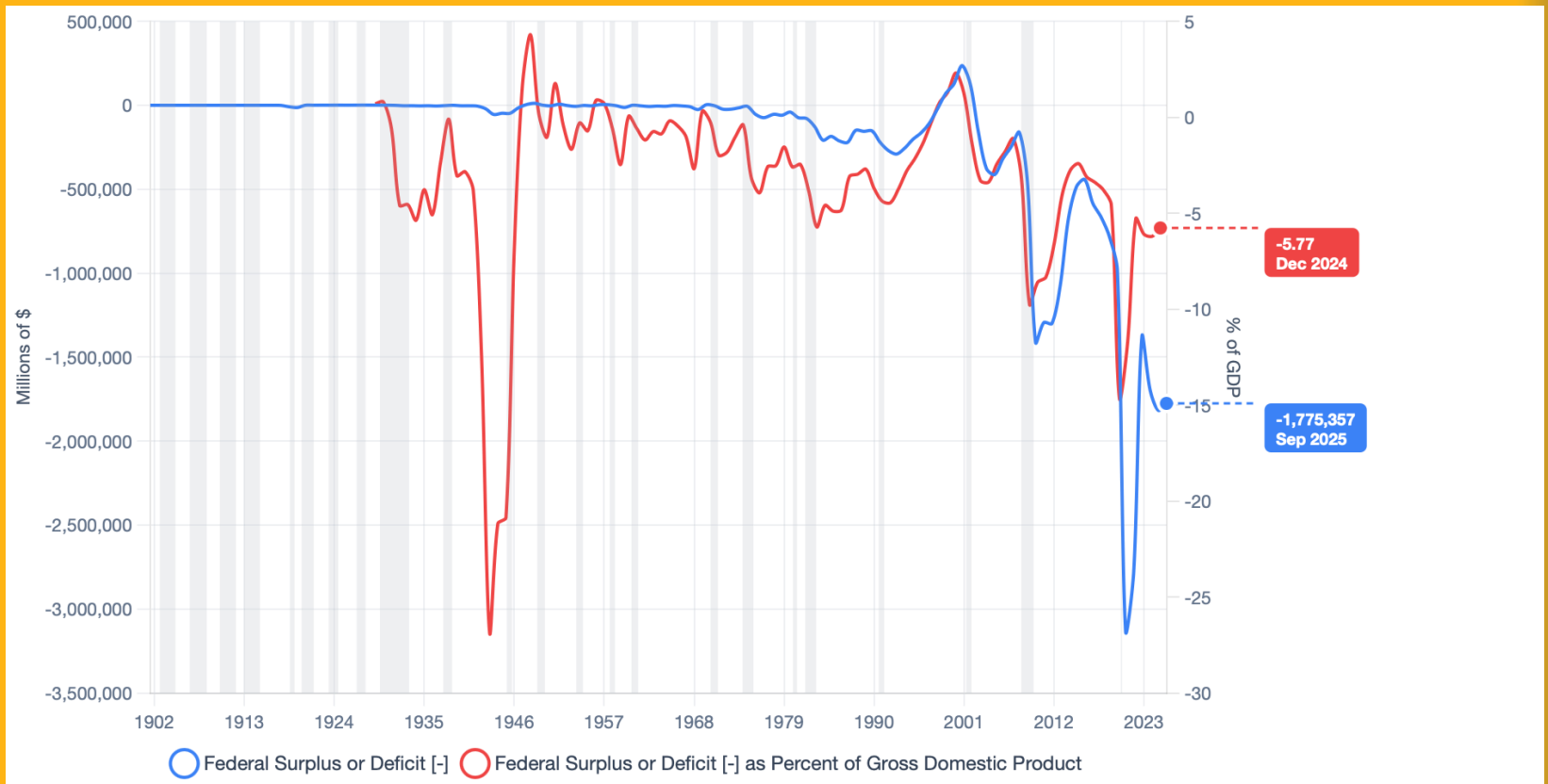
Key Provisions:

- \$203 billion for unemployment benefits
- \$176 billion in tax credits
- \$170 billion for education
- \$174 billion for healthcare

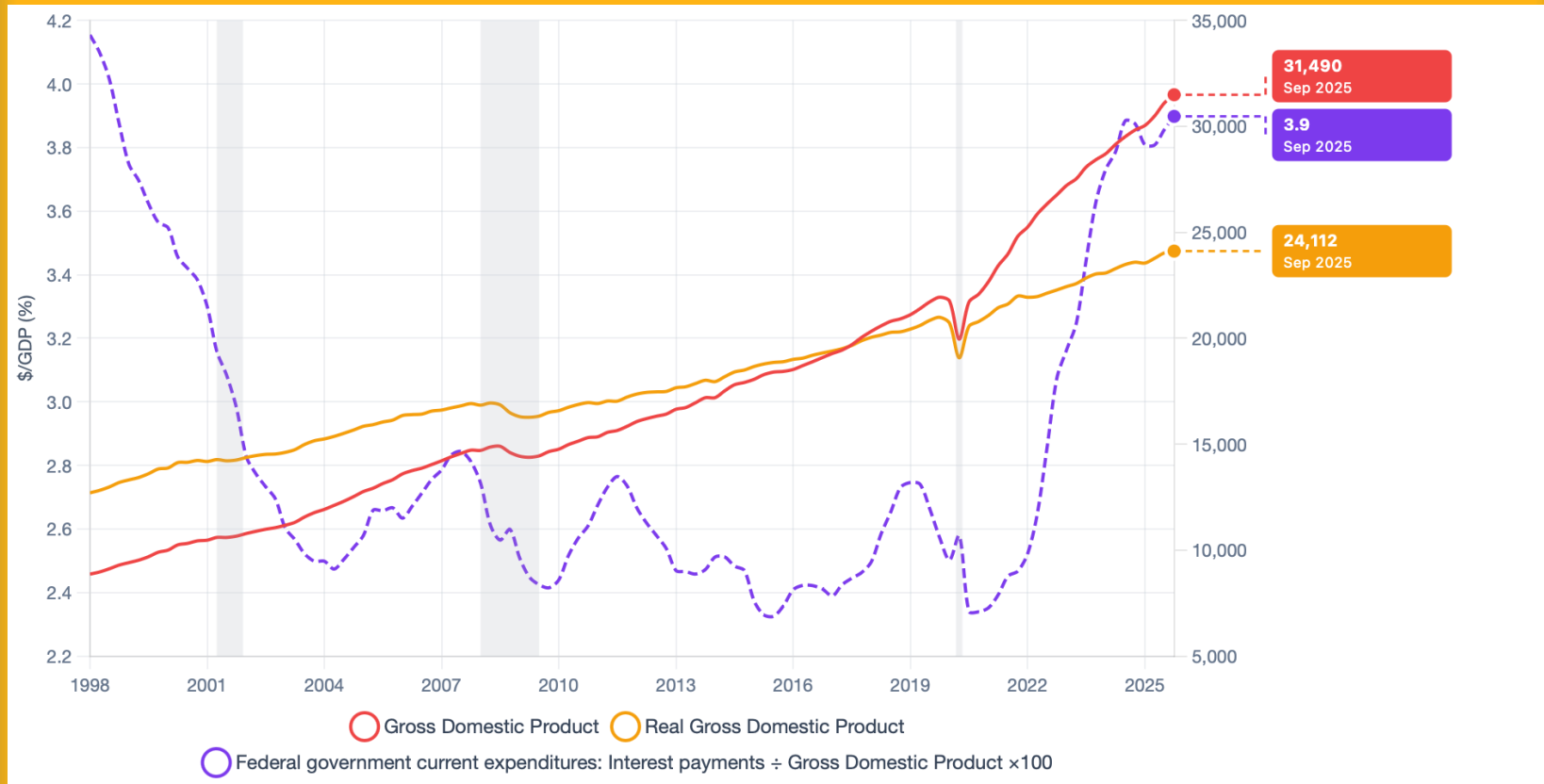
Federal Deficit over Time



Federal Deficit over Time



GDP and Deficit Spending Over Time

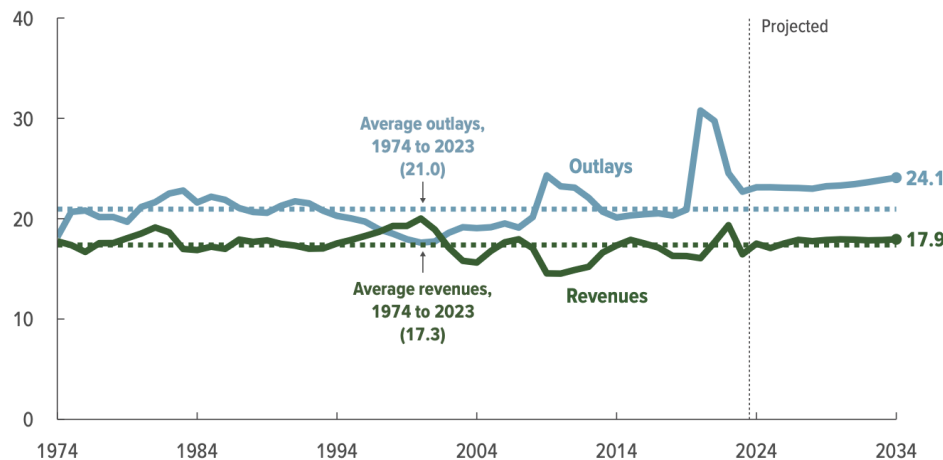


Federal Outlays and Revenues

Figure 1-3.

Total Federal Outlays and Revenues

Percentage of GDP



Measured as a percentage of GDP, projected outlays remain about the same for the next several years as growth in outlays for interest payments is offset by decreases in discretionary spending. Over the 2024–2034 period, outlays exceed their 50-year average by more than revenues exceed their historical average.

Data source: Congressional Budget Office. See www.cbo.gov/publication/59710#data.

When October 1 (the first day of the fiscal year) falls on a weekend, certain payments that would have ordinarily been made on that day are instead made at the end of September and thus are shifted into the previous fiscal year. All projections presented here have been adjusted to exclude the effects of those timing shifts. Historical amounts have been adjusted as far back as the available data will allow.

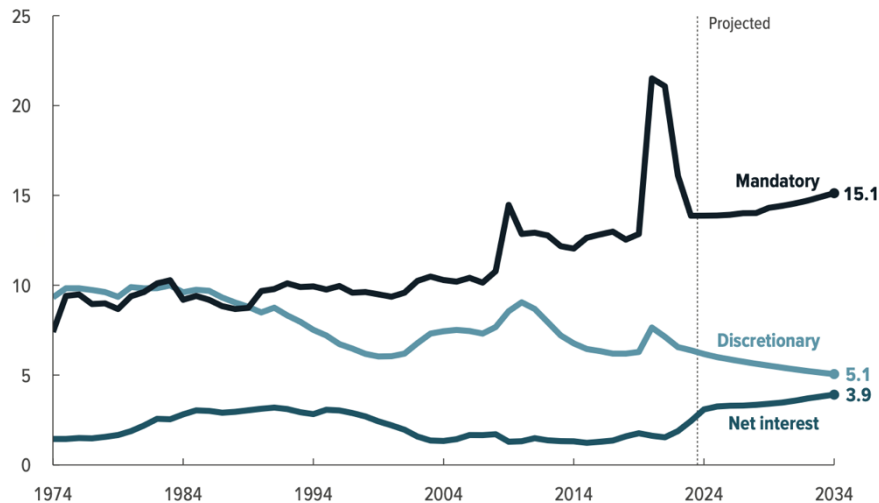
GDP = gross domestic product.

Federal Outlays and Revenues

Figure 1-4.

Outlays, by Category

Percentage of GDP

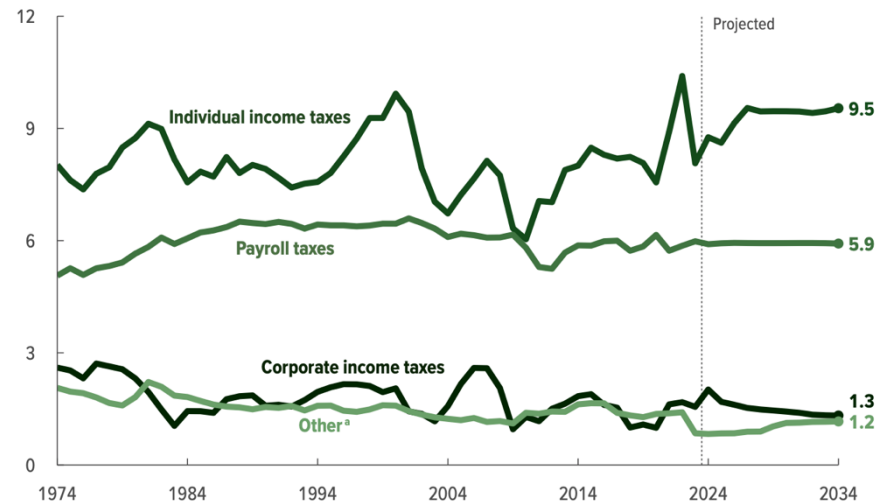


Data source: Congressional Budget Office. See www.cbo.gov/publication/59710#data.

Figure 1-5.

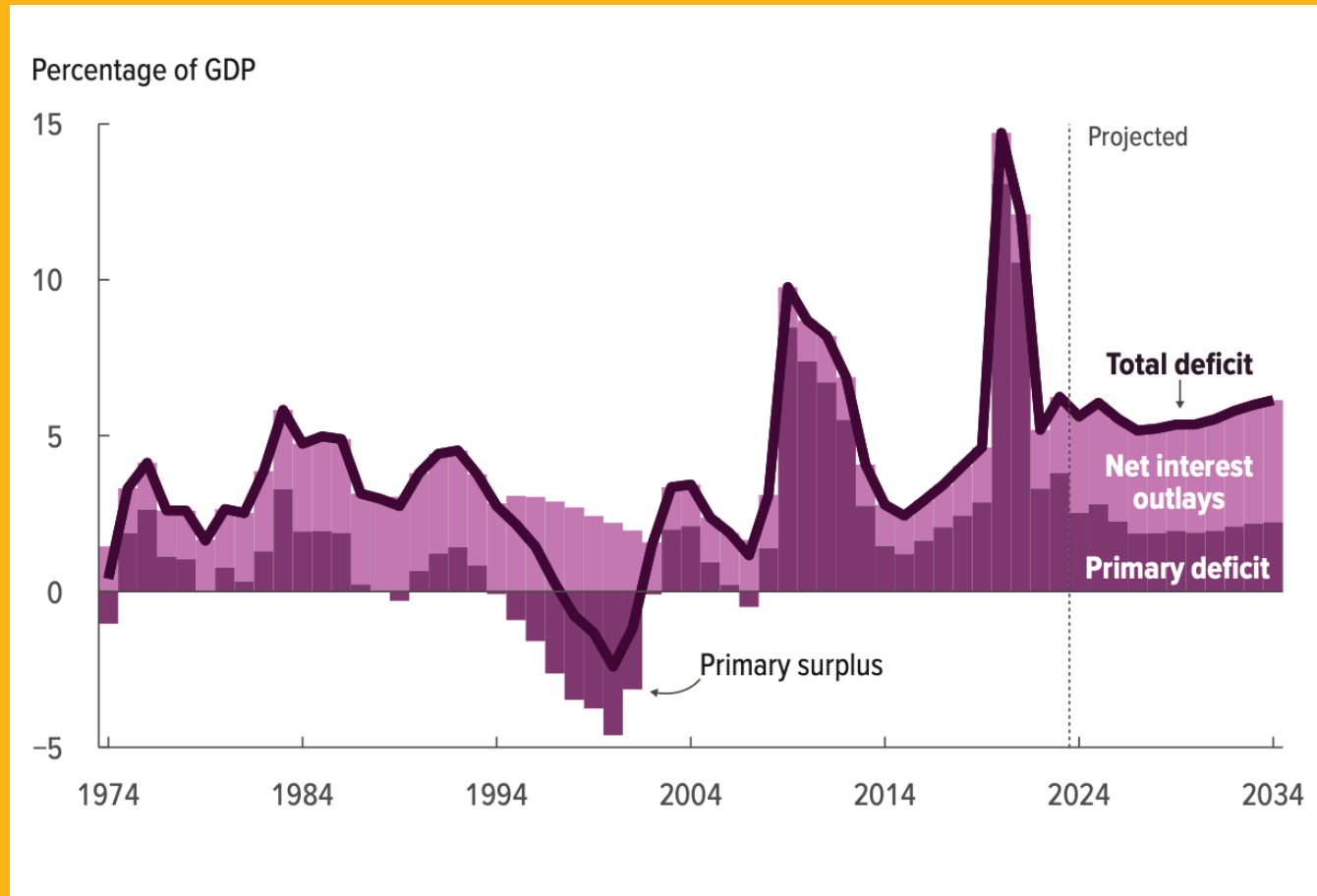
Revenues, by Category

Percentage of GDP

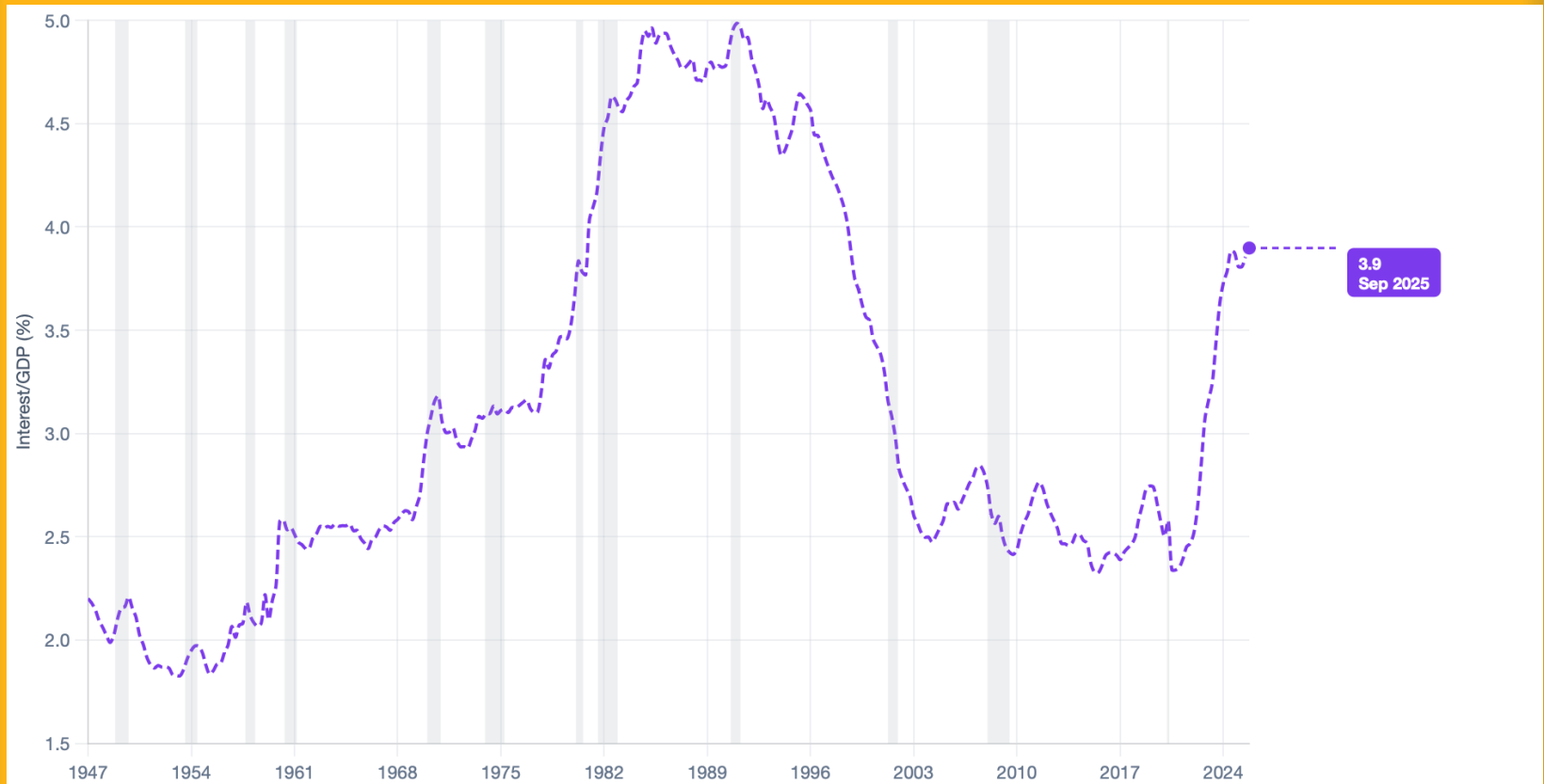


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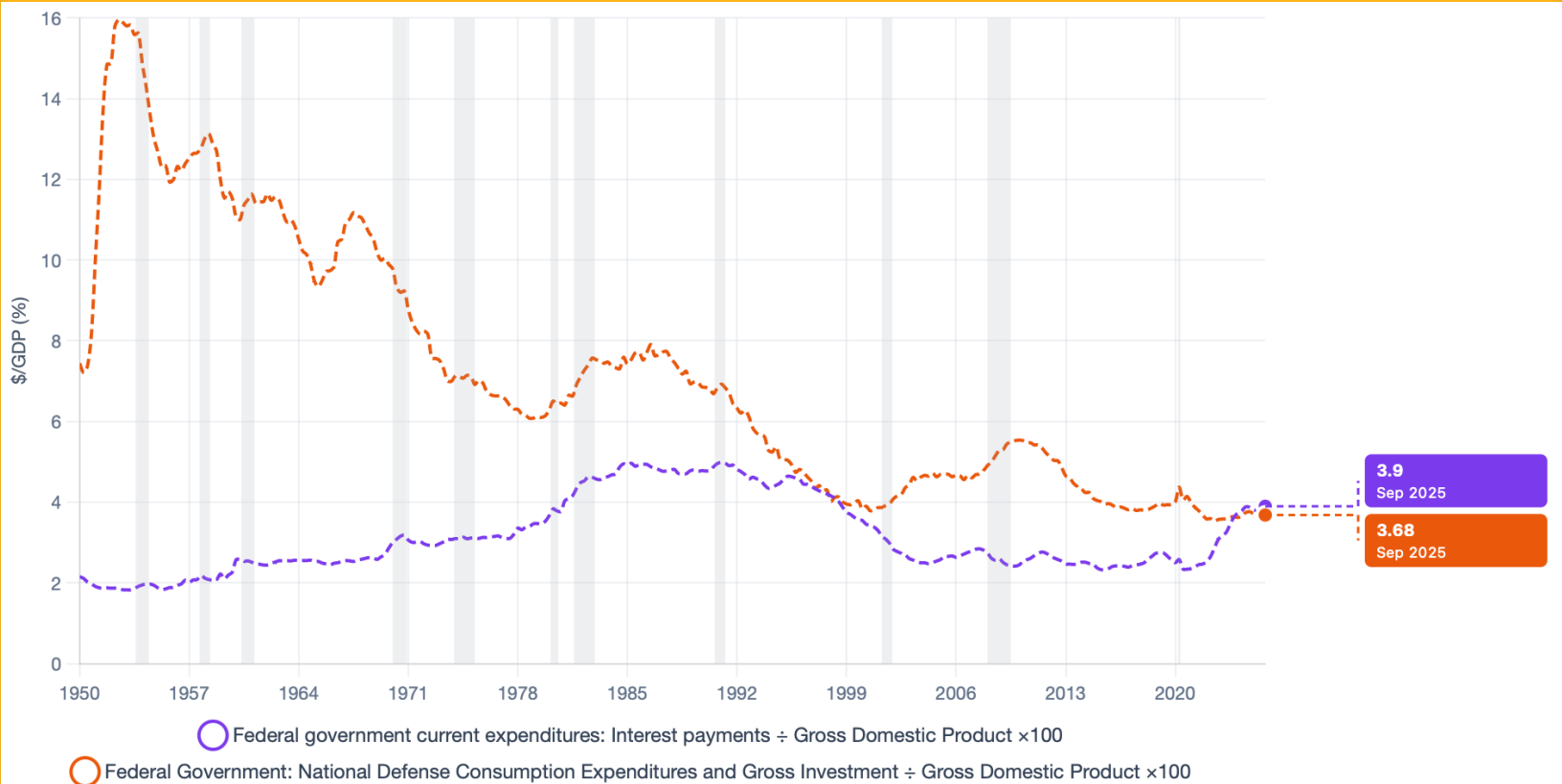
Federal Outlays and Revenues



Net Interest on Debt



Net Interest vs. Defense Spending



Connecting Deficit to Debt

- Each year a deficit is run, money must be borrowed in order to pay for shortfall



How Big is the National Debt?



[Link 1](#)

[Link 2](#)

National Debt Facts

Fact #1: Public Debt vs Intra-Governmental Borrowing

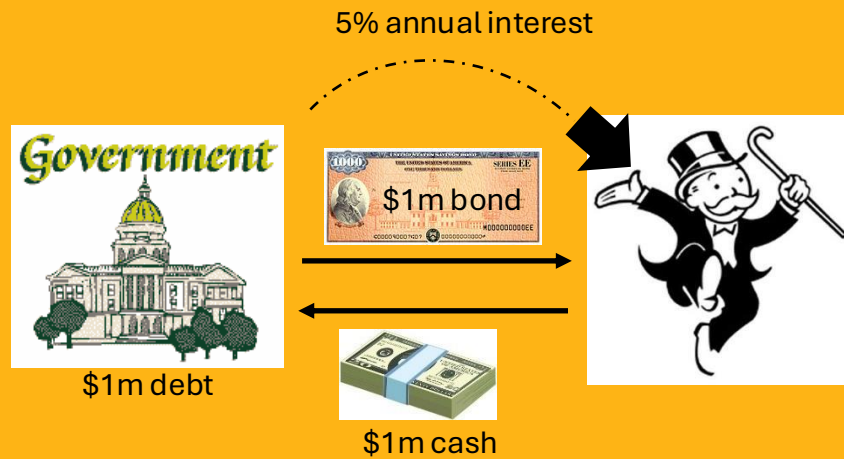


National Debt Facts

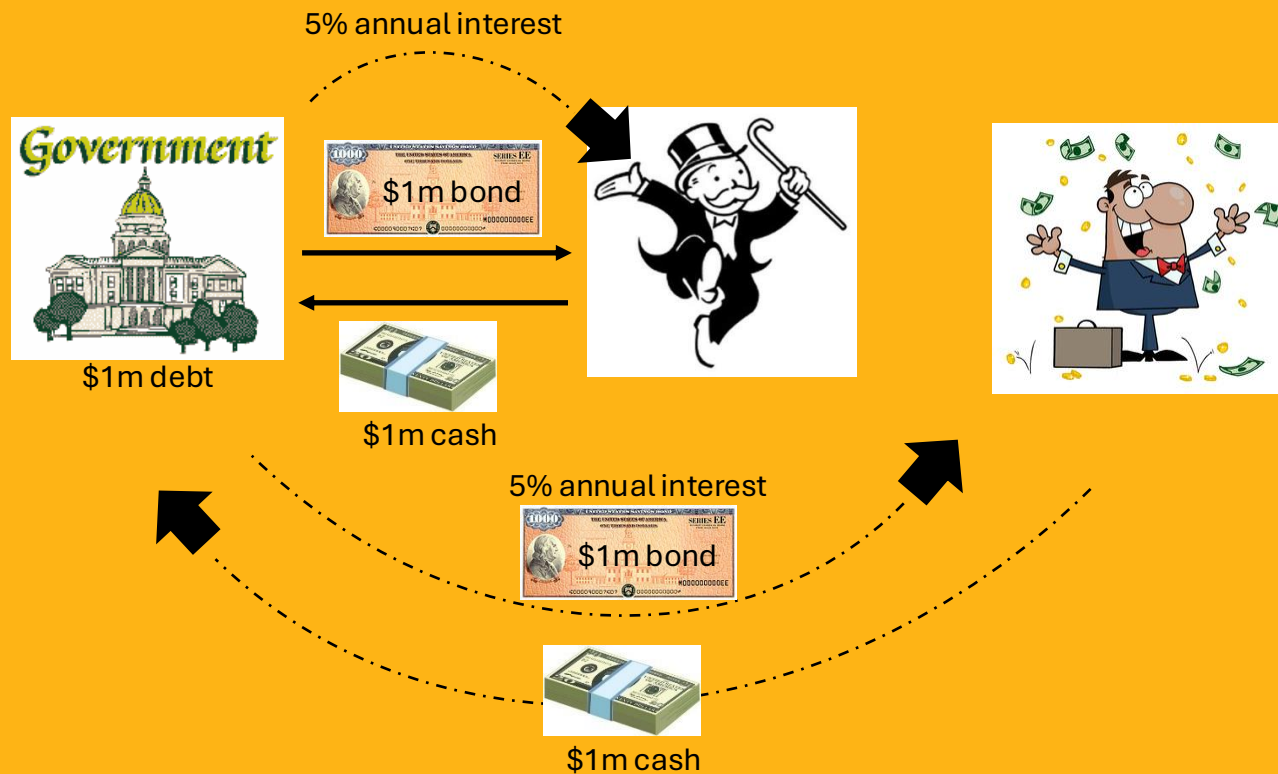
Fact #2: Government Bonds and Rolling Over Debt



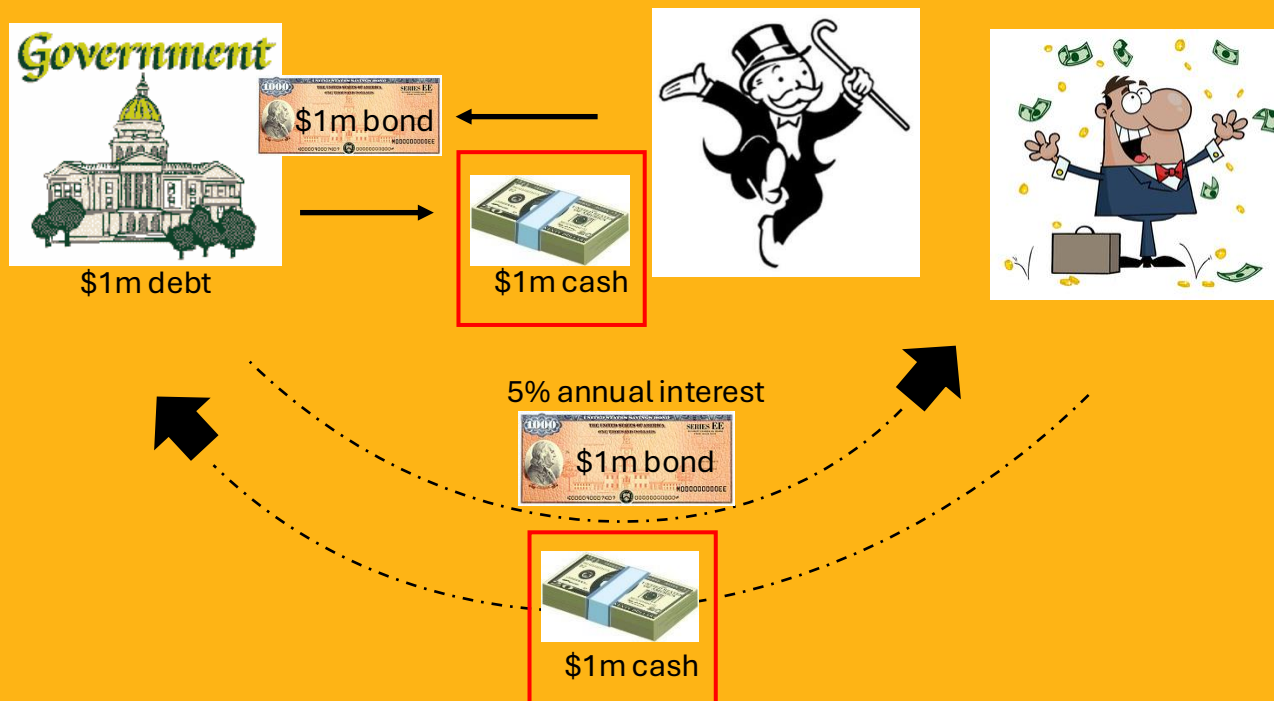
Rolling Over Debt



Rolling Over Debt



Rolling Over Debt



What if the government can't find a lender?



What about the debt matters?

Interest Payments

Example:

Government Debt = 1 million

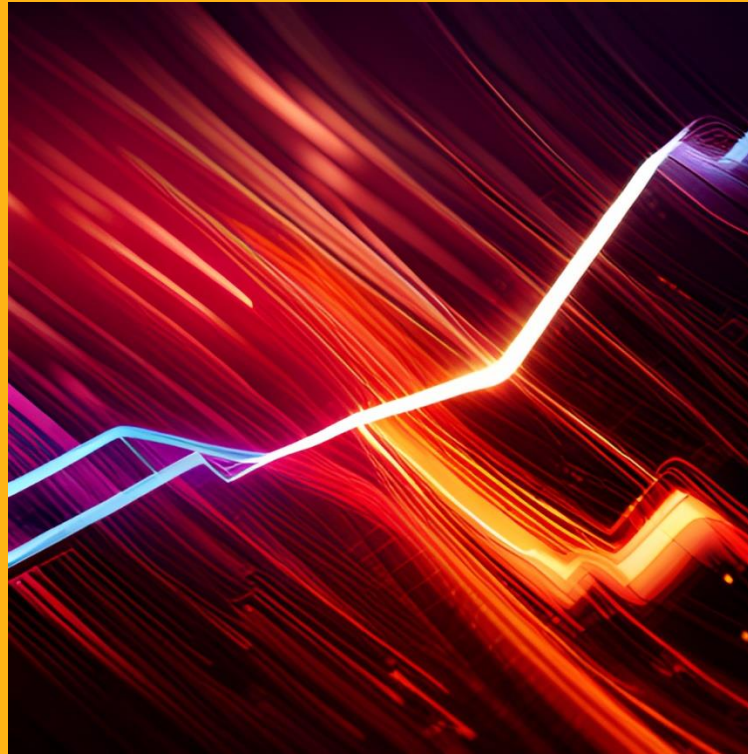
Interest Rate = 5% = 0.05

Nominal GDP = 5 million



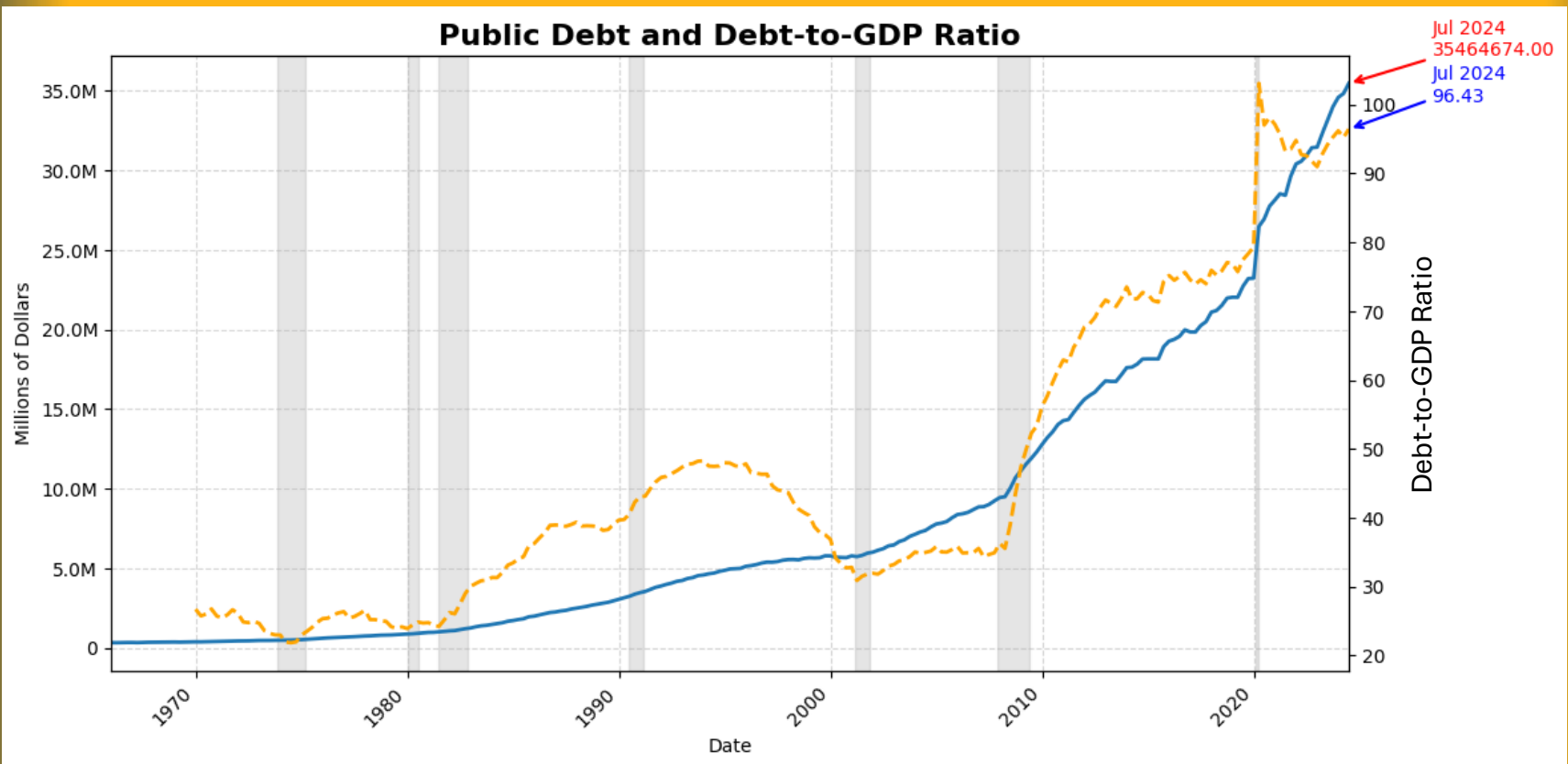
National Debt Facts

Fact #3: Debt-to-GDP Ratio



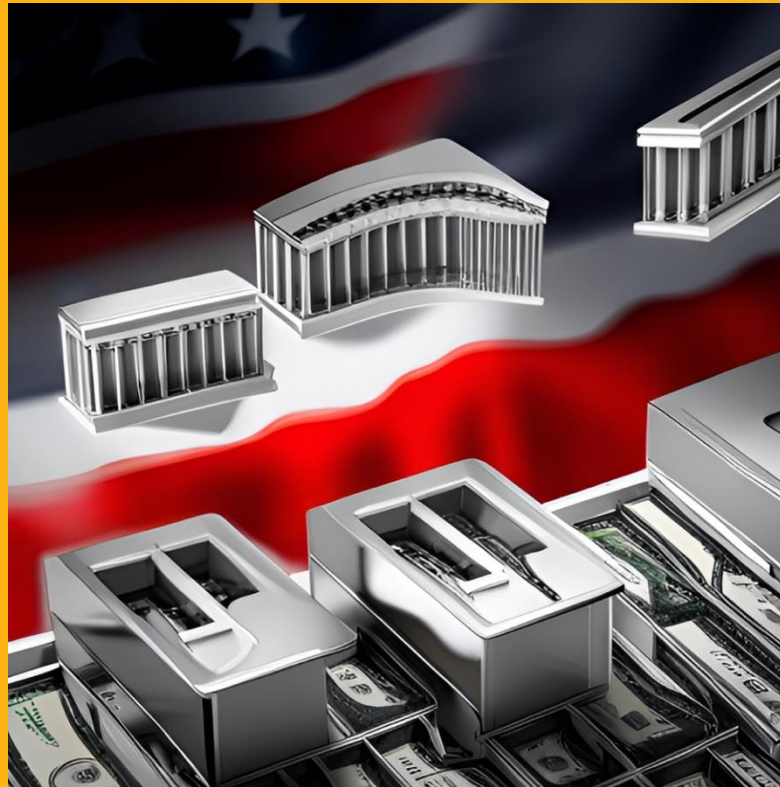
National Debt Facts

- Debt-to-GDP Ratio:

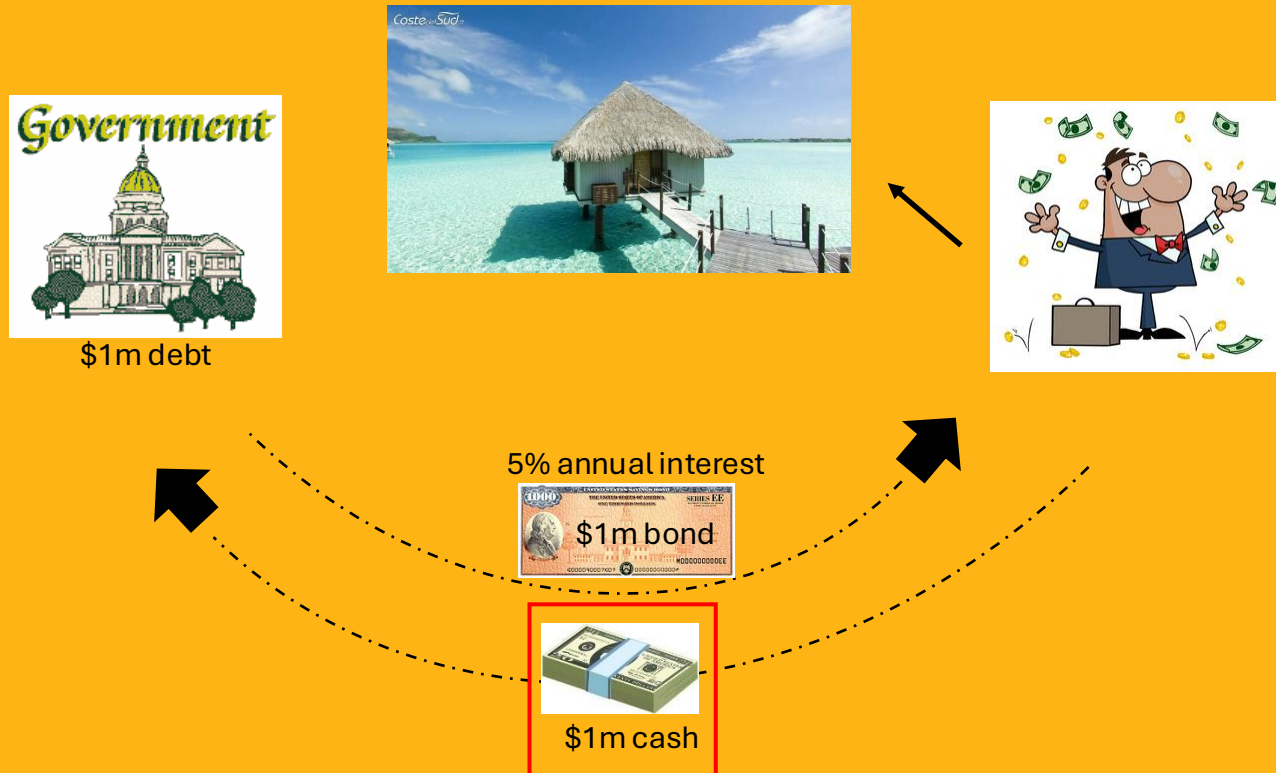


National Debt Facts

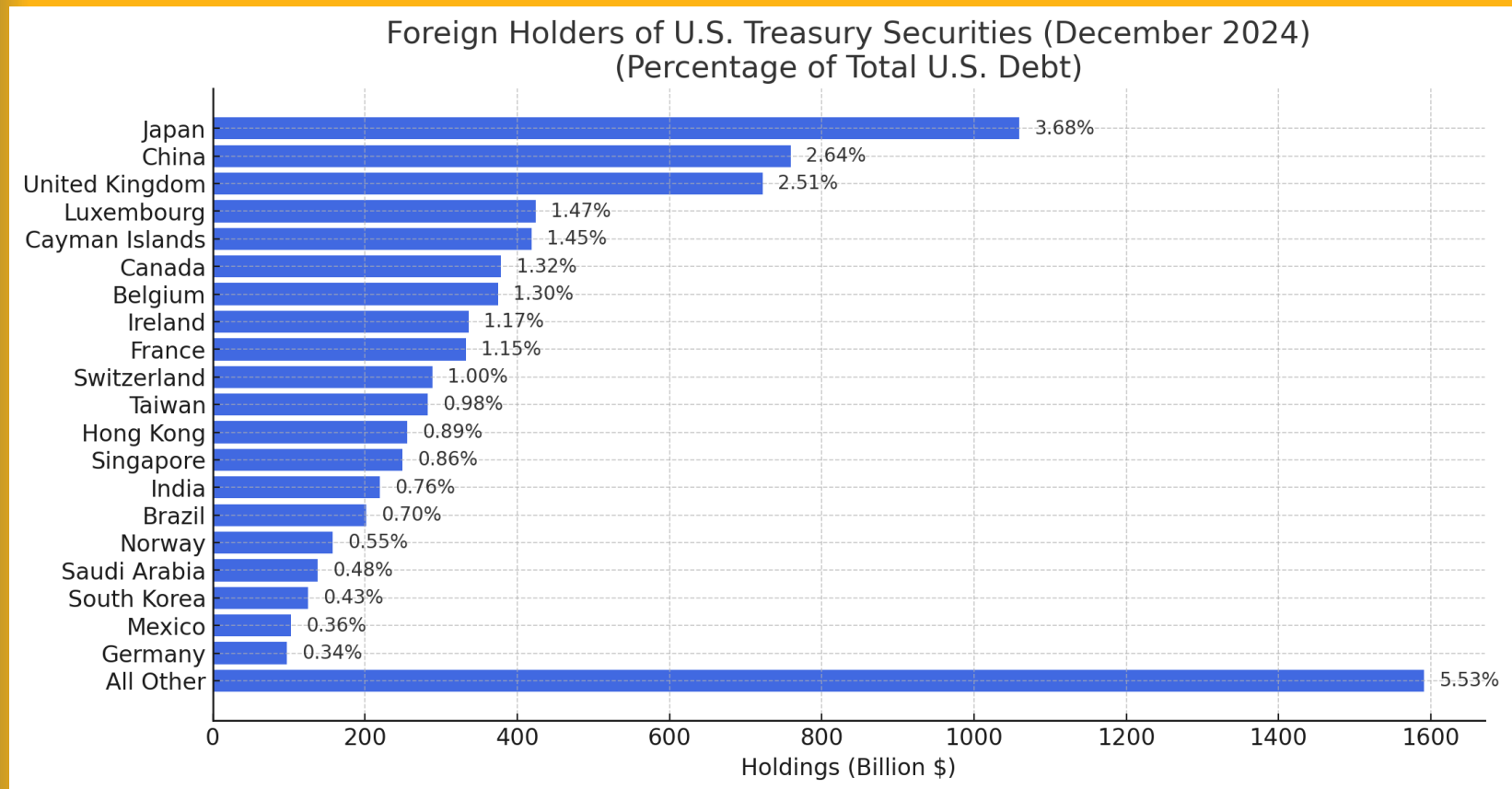
Fact #4: National Debt Interest Payments



Spending Interest Payments



Who Owns the US Debt



Debt Projections and Aggregate Expenditures



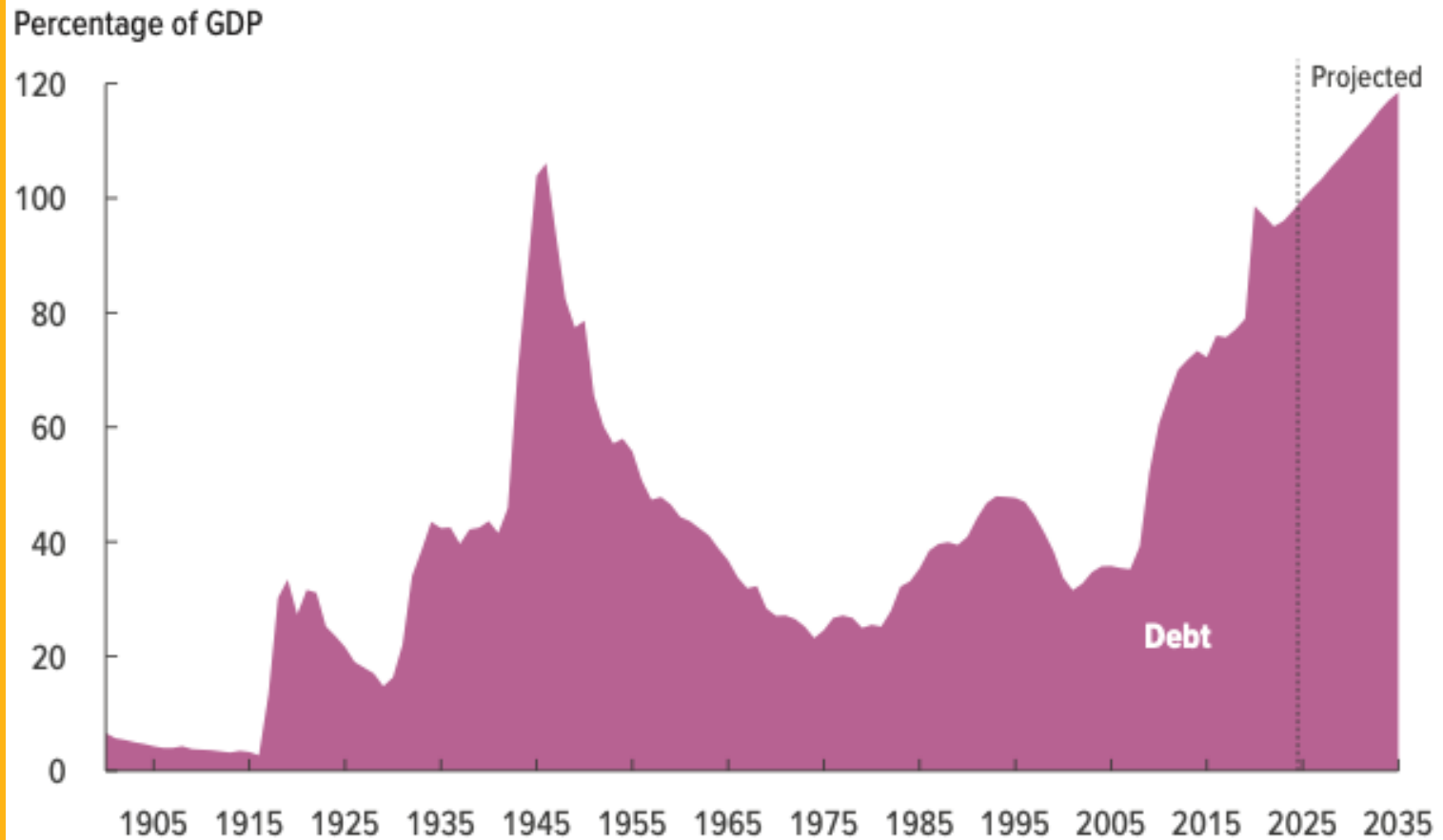
What can be done?

- Decompose the Minimum Tax Rate:

Long Term Costs of Rising Debt

- Fiscal crisis
 - Investors lose confidence in ability to repay its debt/interest
- Higher Interest Payments
 - Larger portion of budget dedicated to interest payments
 - Crowd out other types of government spending
- Economic Impact
 - Serviced and repaid by raising taxes
 - Monetize the debt = inflation, devaluation of currency
- Reduced Flexibility
 - Future generations forced to pay debt of previous generations
 - Limits the resources for economic downturns/emergencies

CBO Debt Projections



Maturing National Debt

- [Average Interest Rate on Debt](#)
- [Unsustainable Fiscal Path, Bureau of the Fiscal Service](#)
- [Fiscal Data \(from Treasury.gov\)](#)

Next Part

- Money
- Federal Reserve
- Monetary Policy
- Connecting Monetary Policy to Aggregate Expenditures



An Economy without Money

Bartering in the 1990s?

“A supervisor at a *fabric factory* here on the outskirts of Moscow, heard good news a couple of weeks ago. Three carloads of *concrete utility poles* had arrived at the train station.”



An Economy without Money

Fabric Factory

Wants: Energy

Has: Fabric

Doesn't Have: Money

Electric Company

Wants: Utility Poles

Has: Energy



Sewing Company

Wants: Fabric

Has: Work Shirts



Auto Manufacturer

Wants: Work Shirts

Has: Car and Truck



Concrete Company

Wants: Car and Truck

Has: Utility Poles

An Economy without Money

Fabric Factory

Wants: Energy

Has: Fabric

Doesn't Have: Money

Energy

Electric Company

Wants: Utility Poles

Has: Energy



Does Bartering Still Exist?

- Barter for a Better Fiji group
- Membership = 190,000 = 20% of Fiji's population
 - pigs for kayaks
 - a violin for a leather satchel
 - doughnuts for building bricks

“I knew that money would be tight to stretch out and even harder to come by. I asked myself what happens when there's no more money? Barter was a natural solution to that.”

- Bartering Time (one hour X for one hour of Y)

Two piglets for a kayak: Fiji returns to barter system as Covid-19 hits economy



▲ Vaziti Masi lost her job due to the coronavirus outbreak, but has been able to keep her flower business afloat

What is Money?

- Three main characteristics:

1. Medium of Exchange:

- Consumers use money to buy things and services
- Merchants accept money as payment due to its convenience in buying products and services
- Bypass the restrictions of a one-on-one bartering system

2. Unit of Account:

- It's a unit for measuring the market value of goods, services, and transactions.
- Conveniently compare product prices and assess their respective worth.
- In a barter system, each exchange has a unique terms-of-trade

3. Store of Value: Unused money today can be stored for future use

- Money can maintain its value over time and withstand inflationary pressures



What is Money?



- **Commodity Money**
 - Good, such as gold or another metal, is used as money
 - Commodity has its own value separate from its use as money
- **Fiat Money**
 - Declared by a government to be legal tender
 - Not backed by an intrinsic value
- **Electronic Money**
 - Only in banking computer systems
 - Increasingly more popular
- **Cryptocurrencies**
 - Digital or virtual currencies that use cryptography for security
 - Not issued by a central authority

Categories of Money

- How much money is in the economy today (money supply)?
- Liquidity: The ability of an asset to be used to purchase a good or service
- Most liquid form of money:
- M1 money supply:
- M2 money supply:
- Credit Cards:
- Moving forward $M1 \text{ money supply} = \text{money supply} = \text{money available today to be used for spending}$



Fractional Reserve Banking

- Bank: a firm that specializes in brokering between savers and borrowers, allowing borrowers to go to one firm instead of 1000s
- Banks are only required to keep a fraction of their deposits on hand as reserves
- Lends out the rest
- Creates new money in the form of a deposit in the borrower's account
- Deposit effectively increases the overall money supply
- Governments regulate banks and provide deposit insurance to protect depositors' funds.



The Federal Reserve



- The Mint Act of 1792 established the dollar as the principal unit of currency
- 1793 to 1861: 1,600+ banks issued their own currency
- 1863: National Banking/Currency Act: National Banking System, Uniform Currency
- Bank Panics in late 1800s/early 1900s: fears that banks would not have funds
- Federal Reserve Act of 1913 Central authority to stabilize system Authorized the issuance of Federal Reserve notes, which are still in use today

The Federal Reserve

- Today there are five key functions of the Federal Reserve:
 1. Overseer of all banks/bank for banks
 2. Bank for the government
 3. Provides loans to banks (discount rate)
 4. Facilitates Issuing of Bonds (auctions)
 5. Carry out monetary policy
- The decisions about monetary policy are carried out at Federal Open Market Committee meetings about 8 times a year (March 18-19)
- Chair: Jerome Powell
6 Board of Governors
12 Independent Branch presidents
- Chair, Board, NY Fed president and rotation of remaining branch presidents vote on policies



Monetary Policy

- Monetary Policy: managing the money supply in order to maximize employment and keep prices stable
- How does the Fed change the Money Supply?
- Typically done through Open Market Operations: buying and selling of Treasury bills (mature in 1 year), notes (2 to 10 years) or bonds (30 years)
- **Open market purchase**→Federal Reserve purchases Treasury securities from banks and the public→more money deposited in banks→**Money Supply Increases!**
- **Open market sale**→Federal Reserve sells Treasury securities that the government has created→less money deposited in banks→**Money Supply Decreases!**

Monetary Policy

- Buying and selling Treasuries changes the interest rate
- Target the Federal Funds Rate: the interest rates banks charge other banks for loans
 - determines flexibility in the banking system since banks can loan out more than they borrow
 - other important interest rates tend to follow the Federal Funds rate